Viral structure and replication cycle

VIRAL STRUCTURE AND REPLICATION (sample notes)

Structure
- Genome: DNA or RNA (single or double stranded, linear or segmented).
- Capsid: protein shell that protects the genome.
- Envelope: lipid membrane acquired from the host cell in some viruses (for example, influenza, HIV). Enveloped viruses are often more sensitive to detergents and drying.

Replication cycle (general)
1. Attachment: viral surface protein binds a host receptor.
2. Penetration / entry: membrane fusion or endocytosis.
3. Uncoating: release of the genome.
4. Replication and transcription: depends on genome type.
5. Assembly: new capsids and genomes are assembled.
6. Release: lysis (non-enveloped) or budding (enveloped).

Note: retroviruses such as HIV use reverse transcriptase to make DNA from their RNA genome, which integrates into the host genome.